HCI: End Semester Exam: Part A (Take Home Exam)

Weightage in End Semester Exam: 50%

Submission Due Date: 13/04/2026

Question 1: Prepare a High Fidelity Prototype of a website or a mobile application and create a video to demonstrate the HCI design for an imaginary product "Time Machine". Assume that this "Time Machine" can help you time travel to any location on the Earth at any time in past or future. Prepare the video that illustrates the sequences of the screens and the user interaction with the screens. Prepare a report as per the following mentioned guidelines. You must work individually and submit your report individually on LMS as per the guidelines. You can NOT use any AI/ML tools or automation tools for this work. [15 Marks]

Following are the essential details user needs to enter in the UI of the Time Machine. These details should be entered on different screens. You may add any intermediate steps in addition to the following essential details needed for building UI/screens. You may add any additional required steps/screens.

- User name and e-mail.
- To which location on the Earth the user wants to travel.
- To which time (date, month and year) user wants to travel (past or future).
- For how much duration (hours and days) user wants to stay at above location.
- For the return journey, at which location user want to return after the mentioned time travel.

Notes:

(1) For every entry by the user, the system (time machine) should respond appropriately and then move forward to the next screen in the sequence to complete all entries.
(2) You should also prepare at-least two screens to highlight any error by the user as illustrations of good error message.
(3) Use the principles of good HCI design learned in the class for above design.

Video 1: Guidelines for Preparing the High Fidelity Prototype demonstration Video:

1. Design all the screens and the user responses using either any UI/UX tool or relevant programming.
2. You must imagine and prepare all the screens where the machine seeks the inputs from the user and how user enters the details. You should imagine the machine's response to the user inputs.
3. Prepare a video that starts from the first screen of your time machine until the end of the user interaction, documenting all the interactions of the user and the machine in the right sequence.
4. You must use your own voice for the required voice-over in the video.
5. Duration of the video should be between 4 minutes to 10 minutes.
6. Save the video in an online folder and include its link on the last page of the report.

Video 2: Guidelines for Preparing the Video to illustrate your proficiency in using the UI/UX tool or programming: The purpose of this video is to illustrate how you have used the UI/UX tool or any programming tool for preparing your High Fidelity Prototype of "Time Machine".

1. Include the following things in the video to demonstrate your proficiency with the tool/programming. Use your voice for "voice-over" and explain the following things.
    a. How you launch the window (or equivalent) for opening the tool/programming.
    b. How you start the new project (or equivalent) in your tool/programming.
    c. How you select the background colour for any one of your screens. Demonstrate changing the colour of the screen with required details.
    d. How you prepare any image/graphics in the tool.
    e. How you can place any diagram/image at a specific location on the screen. Demonstrate moving this diagram/image to another location and describe the technical details.
    f. Demonstrate and explain any two additional features that you find the most significant.
    g. How you connect all the screens. Describe how you can click on one location that moves you to other screen, with details of technicality involved in navigating among different screens.
    h. How you conclude your entire work (Final Execution/error checking if needed etc)
2. Duration of the video should be between 4 minutes to 10 minutes.
3. Save the video in an online folder and include its link on the last page of the report.

Guidelines for Preparing the Report:

1. Prepare a report (MS word or equivalent) and describe the details of each page (screen) you have designed with the photo of each page followed by the description.
2. For each screen, describe its Function, Design Principles/Universal Usability Principles utilized/implemented, details of Interaction Style, if applicable, quality of error message if applicable, and any other technical details (HCI principles/guidelines) relevant to that page (screen) in about 25 to 175 words with required annotations and illustrations.
3. On the first page of the report, clearly mentions your name, enrolment number and relevant details.
4. Store both the videos in any online folder/location. Share the location of the video folders by including their links in the above mentioned report on the last page of the report. Give "view" access of these videos to me and all TAs.
5. Store your entire work online and share the link to your work on the last page of your report.

Guidelines for Submission:

1. Name your report file as HCI_End_Sem_Part_A_2026_< your enrolment number >.
2. Submit your report individually on LMS on or before the end of the day on 13/04/2026.

Evaluation Matrix: [Total 15 Marks]

1. Evaluations based on Video 1 (Video to demonstrate the working of your Prototype):
    a. Completeness of the interaction design to achieve all the goals of the design. [2 Mark]
    b. Quality of Screen designs. [2 Marks]
    c. Sound effects, music (if and when required) and Voice Over for user interactions. [1 Mark].
    d. Creativity and Innovative Approaches in the design. [2 Marks]
2. Evaluations based on Video 2 (Video to demonstrate your proficiency with the tools/programming)
    a. Refer to the guidelines mentioned above for preparing this video. [4 Marks]
3. Evaluation based on Report.
    a. Refer to the guidelines mentioned above for preparing the report. [4 Marks]